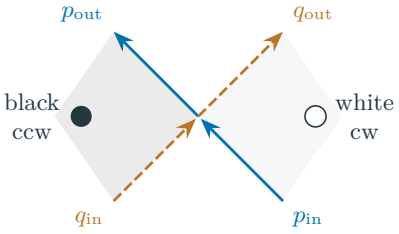
(a)

$L \setminus \{p\} = K \cup \{q\}$

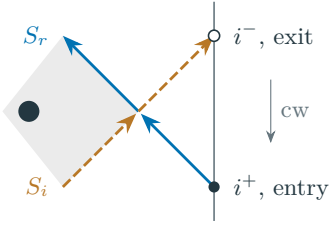


$L \setminus \{q\} = K \cup \{p\}$

black-clique step $p \to q$

$t_p = (-1, 1),\ t_q = (1, 1), \quad \varepsilon_c(p, q) = +1$

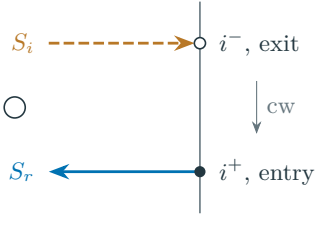
(b)



$r = \bar{f}(i)$: step $r \to i$

one crossing, $\varepsilon_c(r, i) = +1$

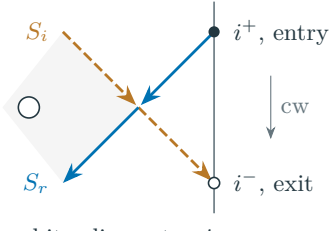
(c)



disjoint connections:

no black corner and no crossing

(d)



white-clique step $i \to r$

one crossing, $\varepsilon_c(r, i) = -1$